

Company-Level Estimates

Cochrane (2005) was the first to estimate a dynamic risk-return model of PE investments taking into account selection bias. He estimates a log-return CAPM, given by

$$\ln(1 + r_i) - \ln(1 + r_f) = \alpha + \beta (\ln(1 + r_m) - \ln(1 + r_f)) + \varepsilon_i, \quad (18.4)$$

where r_i is the return on company i , r_m is the market return, and r_f is the risk-free rate. The coefficients α and β are the log excess return (alpha) and market exposure (beta) of VC investments, respectively. Equation (18.4) is a log version of the standard CAPM regression, which is usually stated in arithmetic returns (see chapters 6 and 10).²⁴

Cochrane finds the effect of selection bias is huge. Leaving aside selection bias, the (log) company alpha is 92%. Taking into account selection bias, the (log) alpha shrinks to -7%. The beta estimate is approximately 2. Portfolio company volatilities are extremely large—over 100% per year. Cochrane finds that performance of VC companies to be very similar to small- and micro-cap NASDAQ stocks. (In fact, the volatilities of comparable listed stocks are even larger than VC portfolio companies.) This reflects the lottery-like nature of starting companies and their high volatility: in PE funds, 85% of returns come from less than 10% of investments and only 13% of investments result in an IPO.²⁵

Morten Sorensen, my colleague at Columbia Business School, finds in Korteweg and Sorensen (2010) that selection bias knocks off 2% to 3% (arithmetic returns) per month from the raw return estimates of VC funds. Korteweg and Sorensen estimate a higher VC beta, around 3, than does Cochrane. They find VC alphas from 1987 to 1993 were positive but modest, alphas during 1994 to 2000 were very high, but the alphas in the 2000s were negative. In particular, post-2001 arithmetic alphas for VC investments have been consistently negative at -3% per month.

Fund-Level Estimates

An advantage of fund-level estimates is that since PE funds have similar lifetimes (around 10 years) and selection issues are smaller, analysis can be carried out in more conventional arithmetic, rather than log returns. Using fund-level cash flows, Phalippou and Gottschalg (2009) find that PE funds have underperformed the S&P 500 by 3%. After taking into account additional risk factors to capture the size premium and the value-growth premium (see chapter 7), the risk-adjusted underperformance is 6%.

²⁴ The alpha in equation (18.4) is not a holding-period excess return. That is, there is a big difference between arithmetic and continuous-time (or log) returns, and that difference is due to volatility. The arithmetic alpha is approximately $\alpha + \frac{1}{2}\sigma^2$, where σ is the volatility of returns (see the Appendix).

²⁵ See Ewens and Rhodes-Kropf (2012).